

AUTOSAR BODY CONTROL SYSTEM

High-Level Design Document

Document ID: HLD-BCS-001

Revision: 1.0

Status: Approved for Engineering Analysis

Classification: Synthetic Demonstration Data

1. System Overview

The Body Control System (BCS) coordinates body-related vehicle functions including door state monitoring, window commands, and vehicle state information. The architecture follows an AUTOSAR software-component model in which software components communicate through typed interfaces and ports.

2. Software Architecture

The primary software components are BodyControlManager, DoorControl, WindowControl, and VehicleStateManager. BodyControlManager acts as the central coordination component. DoorControl publishes door state information. WindowControl consumes window commands. VehicleStateManager provides vehicle state information such as vehicle speed.

3. Software Components

Component	Role	Primary Responsibility
BodyControlManager	Coordinator	Coordinates body-control functions and consumes door and vehicle state information.
DoorControl	Provider	Monitors door status and publishes door-state information.
WindowControl	Consumer	Controls window operation based on commands and vehicle state.
VehicleStateManager	Provider	Provides vehicle-level state information including vehicle speed.

4. Interfaces

Interface	Provider	Consumer	Signals
IDoorStatus	DoorControl	BodyControlManager	DoorOpenStatus
IWindowCommand	BodyControlManager	WindowControl	WindowPosition
IVehicleState	VehicleStateManager	WindowControl	VehicleSpeed

5. Ports

Component	Port	Direction	Interface
DoorControl	DoorStatus_Out	P-PORT	IDoorStatus
BodyControlManager	DoorStatus_In	R-PORT	IDoorStatus
BodyControlManager	WindowCommand_Out	P-PORT	IWindowCommand
WindowControl	WindowCommand_In	R-PORT	IWindowCommand
VehicleStateManager	VehicleState_Out	P-PORT	IVehicleState
WindowControl	VehicleState_In	R-PORT	IVehicleState

6. Signals

Signal	Data Type	Source	Destination
DoorOpenStatus	boolean	DoorControl	BodyControlManager
WindowPosition	uint8	BodyControlManager	WindowControl
VehicleSpeed	uint16	VehicleStateManager	WindowControl

7. Dependencies

Source	Relationship	Target	Reason
DoorControl	PROVIDES_INTERFACE	IDoorStatus	Publishes door state.
BodyControlManager	REQUIRES_INTERFACE	IDoorStatus	Consumes door state.
BodyControlManager	PROVIDES_INTERFACE	IWindowCommand	Generates window commands.
WindowControl	REQUIRES_INTERFACE	IWindowCommand	Consumes window commands.
VehicleStateManager	PROVIDES_INTERFACE	IVehicleState	Provides vehicle state.
WindowControl	REQUIRES_INTERFACE	IVehicleState	Uses vehicle speed.

8. Functional Flows

Flow F-001: Door State Monitoring

DoorControl reads the physical door state and publishes DoorOpenStatus through IDoorStatus.

BodyControlManager receives the signal through DoorStatus_In and updates the body-state model.

Flow F-002: Window Command

BodyControlManager evaluates the requested window operation and publishes WindowPosition through IWindowCommand. WindowControl receives the command through WindowCommand_In.

Flow F-003: Vehicle State Dependency

VehicleStateManager publishes VehicleSpeed through IVehicleState. WindowControl consumes VehicleSpeed to apply safety-related window-operation constraints.

9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

10. Engineering Traceability

This synthetic HLD is intentionally structured to support document extraction, semantic retrieval, relationship mapping, validation, revision comparison, and evidence-based question answering.